

Agenda

- 1) Keys
- 2) Introduction to SQL

Keys

→ Every row should be unique in Relational DB

1) Super Key

roll no.	name	year	01/06	02/04	03/06
	Abhishek	1995	Y	Y	Y
	Abhishek	1995	N	N	N
	⋮				
	Shridhar	1995	-		

Key:- A col or a set of cols of a table whose values are guaranteed to be able to identify a row uniquely.



Super key:- A col or a set of cols of a table whose values are guaranteed to be able to identify a row uniquely.

(phone no.) → ✓ } Super Keys

(email) → ✓ }

(name) → X

(psp) → X

(batch) → X

(phone no, email) → ✓

(phone no, name) → ✓

(email, name) → ✓

(phone no, psp) → ✓

Super Keys

(phone no, psp, batch) → ✓ } Super Keys

(phone no, batch) → ✓

(psp, batch, name) → X

(psp, batch) → X

Students

name	psp	email	batch	phone number
Alok	95		2	
Saloni	95		2	

Super keys → Coln or set of columns that identified every row uniquely.

eg. Students.

name	psp
Alok	95
Alok	95

If it is not guaranteed that every name in this table will be unique, then there are 0 keys.

→ Violates Relational Model.

Quiz

{ Student Id, Course Id } → Yes

{ FN, LN } → X

{ Age, Course Name } → X

{ LN, Course Id } → X

Candidate key

{email}

{email, psp}

{phone no.}

{phone no., batch-id}

{phone no., batch-id, psp}

} Super Keys

* Candidate keys are super keys of smallest size.

* Super key where all the cols are contributing to help uniquely identify a row in a table.

(~~name~~, phone, ~~psp~~) $\rightarrow$ (phone) $\rightarrow$ candidate key

(phone, psp) $\rightarrow$ X Not a candidate key

Attendance.

PK = (class-id, stu-id)

stud-id	class-id	attendance-percent	any other col
400	191	87%	
401	191	40%	
402	191	30%	
400	202	40%	
401	210	80%	
402	210	80%	

Super Keys.

(student-id) X

(class-id) X

(attendance) X

(student-id, attendance) X

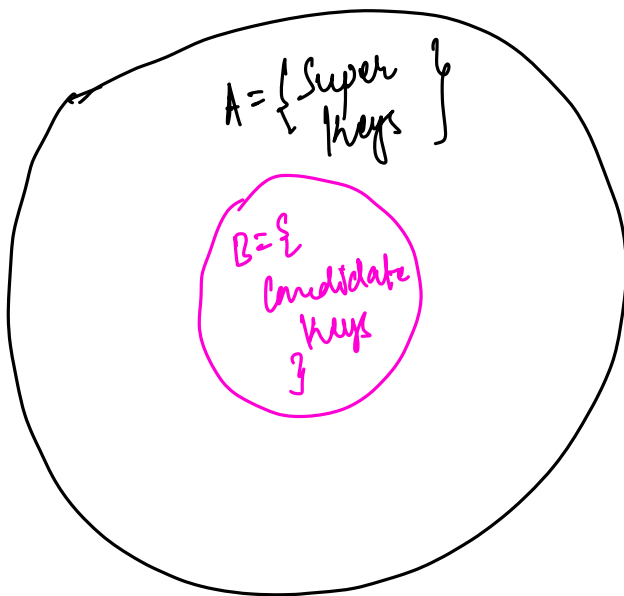
(class-id, attendance) X

(student-id, class-id) ✓ → Candidate Key

(student-id, class-id, attendance) ✓ →

(student-id, class-id, any other col.) ✓

Candidate Key:- Super Keys where each col is mandatory to uniquely identify every row.



Super Keys:- Party Workers (100)
Candidate Keys:- Best Super Keys (10)
Primary Key:- Chief Minister (1)

→ Any database table can have a lot of Candidate Keys.
(email) (phone no.)

→ In reality, when you create a table in SQL Database, it forces us to specify a set of columns that will be able to uniquely identify every row.

only 1

Primary Key :- A candidate Key selected to help identify each row of the table uniquely.

Why?

→ Database stores data internally sorted by the PK.

e.g. Table with student data
(1M students) (1M rows)

PK $\Rightarrow$ (email)

name	age	batch-id	prp	email
Alok	25	1	99	5000
Sahil	22	3	99	
.	.	.	.	
Vishal	23	1	95	1000
.	.	.	.	
.	.	.	.	
Mounika	25	1	99	1 lakh
.	.	.	.	

3 5000 students.

7 1000 students

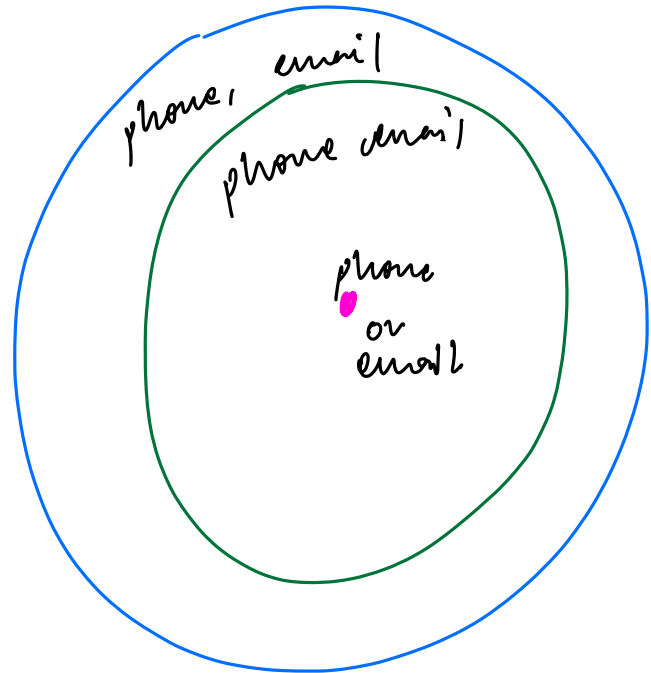
} 1 latch

→ Database creates an index on the PK
→ Later

□ → Superkeys

□ → Candidate keys.

□ → Primary key



Break till:- 10:38 PM

employees.
empId → Primary key

1		
2		
4		
9		
100		
101		

Select * from employees.

You will get data sorted by employee-id.

What is a good primary key?

- 1) Should be fast to sort on. $\Rightarrow$ prefer to have PK with single integer col. (Integer col)
- 2) Smaller size (single col) $\rightarrow$ Indexes.
- 3) Should never change (email X, phone X, address X)

(Integer)

id	name	email	phone No.	batch-id
1				
2				
3				

PK column

Databases support having an auto-increment id column.

(Candidate key)

	SK	CK	PK
name	X	X	X
email	✓	✓	Have it, but not advised.
ph. no.	✓	✓	- - -
batch-id	X	X	X
id	✓	✓	✓

Use this in 99% of cases.

attendance

id	class-id	student-id	attendance.
1	1	Alok	99
2		Mukul	
3		Neha	
4	2	Alok	99
5		Lushanifa	

Composite key

⇒ A Key with more than 1 col.

↓
uniquely identifies every row.

Superkey :-
(phone no.) → X
(email) → X
(id) → X
(phone, email) → composite super key.

(phone, email, name) → composite super key.

attendance :-

id	stu-id	class-id	attendance percent
----	--------	----------	--------------------

PK :- id → Normal

OR

stu-id | class-id → Composite PK

↳ composite key → Also a composite candidate key

Revise All Keys.

↳ After all keys, we will do a revision